

NutriFLEX

EXECUTIVE SUMMARY

NutriFLEX is an AI-powered nutrition assistant designed to offer personalized solutions in nutrition management. The app leverages advanced machine learning algorithms to predict calorie content, assist in meal planning, and recommend recipes tailored to individual health needs. Built on a Streamlit framework, NutriFLEX provides a user-friendly interface for ease of use and interaction.

Key functionalities include:

- **BMR and TEE Calculation:** The app calculates Basal Metabolic Rate (BMR) and Total Energy Expenditure (TEE) to guide users in creating personalized meal plans.
- **Calorie Prediction:** Using machine learning models like Random Forest, XGBoost, Gradient Boosting, and CatBoost, NutriFLEX accurately predicts the calorie content of various foods.
- **Recipe Recommendations:** Users can receive both traditional and AI-generated smart recipe suggestions based on their nutritional goals and preferences.
- **Gamification:** The app gamifies the nutrition experience by rewarding healthy food choices, encouraging long-term engagement.
- **Health Goal Tracking:** NutriFLEX dynamically adjusts meal suggestions as users progress toward their personalized health goals.

2. INTRODUCTION

In an era where personalized nutrition is increasingly recognized as essential to maintaining overall health and preventing lifestyle-related diseases, NutriFLEX stands as a comprehensive AI-driven solution. It simplifies nutrition management by offering an interactive platform where users can manage their dietary intake more effectively.

The NutriFLEX app integrates several advanced features:

1. **Calorie Prediction:** With the help of machine learning models, NutriFLEX provides precise calorie predictions for a wide range of food items, enabling users to make informed dietary choices.

2. **Meal Planning:** By using BMR and TEE calculations alongside individual dietary preferences, the app recommends balanced meals that align with the user's nutritional requirements.
3. **Gamification:** To foster better engagement, NutriFLEX rewards users for making healthy choices, which helps to encourage consistent behavior change.
4. **Recipe Recommendation:** NutriFLEX offers both traditional recipes and innovative AI-generated suggestions, ensuring that users have access to creative, healthy, and goal-oriented meal options.

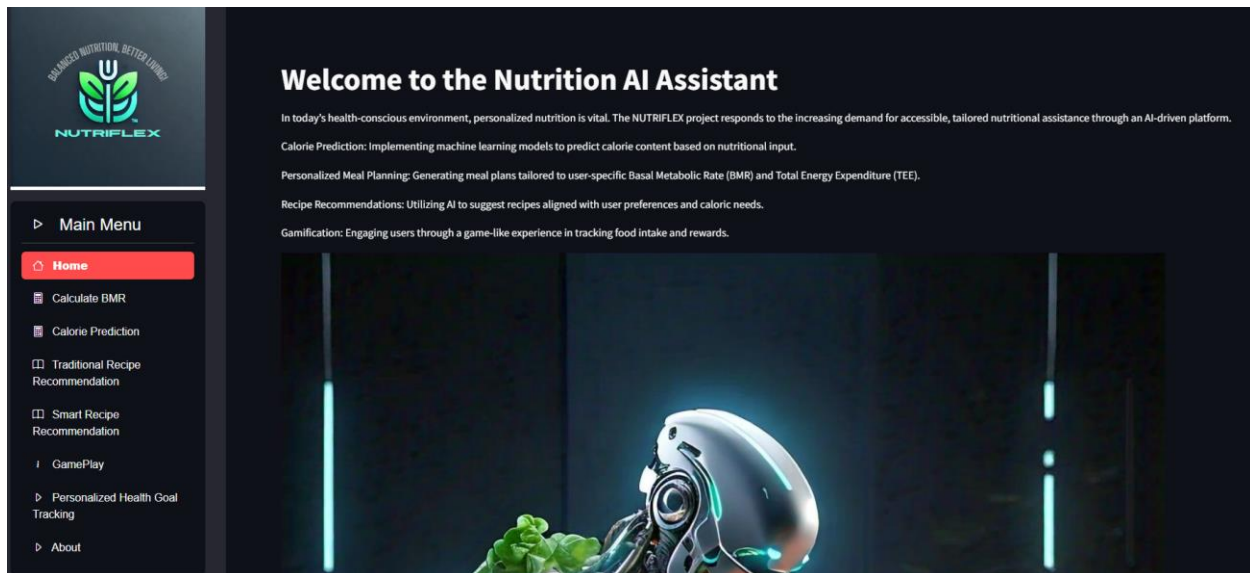


Figure 2: - INTRODUCTION TO NutriFLEX

3. METHODOLOGY

This section explains the techniques used to build the NutriFLEX app.

3.1 DATA PREPROCESSING

The nutrition dataset is preprocessed to remove irrelevant columns and handle missing values. Categorical columns such as FoodGroup and ScientificName are one-hot encoded, while numerical values like calories and macronutrients are scaled using StandardScaler.

3.2 CALORIE PREDICTION MODELS

Four machine learning models were implemented:

- Random Forest
- XGBoost
- Gradient Boosting
- CatBoost

Each model is trained to predict the calorie content based on nutritional inputs such as protein, fat, and vitamins. The models are pre-trained and saved as pickle files, which are loaded at runtime in the app.

The screenshot shows a web application titled "Meal Planner based on Physique". It features a dark theme with white text. The interface includes several input fields for user data: "Enter your age:" with a value of 34, "Select gender:" with a dropdown menu showing "Male", "Enter weight (in kg):" with a value of 68.00, "Enter height (in cm):" with a value of 175.54, and "Select activity level:" with a dropdown menu showing "lightly active". Below these fields is a red-outlined button labeled "Generate Meal Plan". Underneath the button, the application displays calculated values: "Your BMR: 1612.12 kcal" and "Your TEE: 2216.67 kcal". At the bottom, there is a section titled "Meal Plan based on your TEE:" which contains a table with one row of data.

	ShortDescrip	Energy_kcal	Protein_g	Fat_g	Carb_g
0	BUTTER,WITH SALT	717	0.85	81.11	0.06

Figure 3.3: - BASAL METABOLIC RATE (BMR) AND TOTAL ENERGY EXPENDITURE (TEE)

3.3 BASAL METABOLIC RATE (BMR) AND TOTAL ENERGY EXPENDITURE (TEE)

The Mifflin-St Jeor equation is used to calculate BMR, which is then adjusted for activity levels to calculate TEE. These values guide meal plan recommendations based on energy requirements.

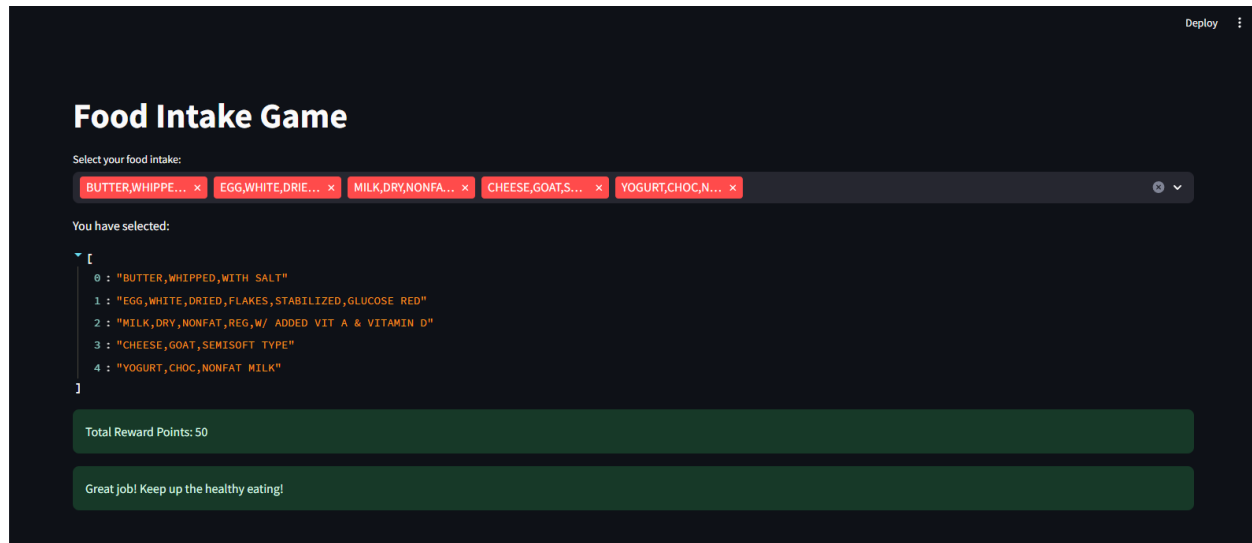


Figure 3.4: - GAMIFICATION AND REWARD SYSTEM

3.4 GAMIFICATION AND REWARD SYSTEM

The app includes a "Food Intake Game" that awards points based on the selection of healthy food items. This gamification aspect encourages users to engage more with the app and make healthier choices.

3.5 RECIPE RECOMMENDATION

The app provides two types of recipe recommendations:

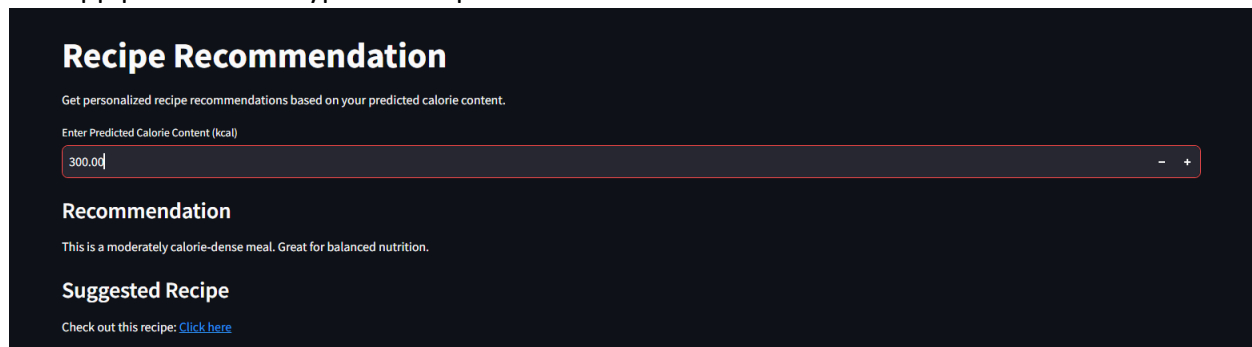


Figure 3.5.A: -TRADITIONAL RECOMMENDATIONS

- **Traditional Recommendations:** Based on calorie thresholds, suggesting predefined recipes from a list of online sources.

Smart Recipe Recommendation

Enter your desired calorie count:

300

Generate Recipe

Spinach & Broccoli Chicken Stir-Fry (Approx. 300 calories)

This quick and easy stir-fry packs a punch of flavor and nutrients in under 300 calories.

Ingredients:

- 1 boneless, skinless chicken breast (approx. 150g), cut into bite-sized pieces
- 1 tablespoon olive oil
- 1 cup broccoli florets
- 1 cup spinach
- 1 clove garlic, minced
- 1/4 cup low-sodium chicken broth
- 1 tablespoon soy sauce
- 1 teaspoon sesame oil
- Salt and pepper to taste
- Optional: 1/4 cup brown rice or quinoa for serving

Instructions:

1. **Prep:** In a bowl, toss chicken with 1/2 teaspoon salt and 1/4 teaspoon black pepper. Set aside.
2. **Cook Chicken:** Heat olive oil in a large skillet over medium-high heat. Add chicken and cook until browned and cooked through, about 5-7 minutes. Remove chicken from pan and set aside.
3. **Sauté Vegetables:** Add broccoli florets to the skillet and cook for 2-3 minutes, stirring occasionally, until slightly tender.

Figure 3.5.B: - SMART RECOMMENDATIONS

- **Smart Recommendations:** Using AI, smart recipes are generated dynamically based on user input. This feature allows users to receive custom recipes with calorie targets, macronutrient breakdowns, and preparation steps.

Enter your target daily calorie intake:

300

Meal Suggestions Based on Your Intake

	ShortDescrip	Energy_kcal	Protein_g	Fat_g	Carb_g
11	CHEESE,COTTAGE,CRM	98	11.12	4.3	3.38
12	CHEESE,COTTAGE,CRM	97	10.69	3.85	4.61
13	CHEESE,COTTAGE,NON	72	10.34	0.29	6.66
14	CHEESE,COTTAGE,LOW	81	10.45	2.27	4.76
15	CHEESE,COTTAGE,LOW	72	12.39	1.02	2.72
55	EGGNOG	88	4.55	4.19	8.05
57	MILK,FILLED,FLUID,W/I	63	3.33	3.46	4.74
58	MILK,FILLED,FLUID,W/I	63	3.33	3.4	4.74
68	MILK SUBSTITUTES,FLI	61	1.75	3.41	6.16
69	MILK,WHL,3.25% MILKI	61	3.15	3.25	4.8

Figure 3.6: - PERSONALIZED HEALTH GOAL TRACKING

3.6 PERSONALIZED HEALTH GOAL TRACKING

The app allows users to set personal health goals and tracks progress over time. Based on the user's current food intake, it suggests appropriate meals to meet or adjust their caloric intake.

4. USER INTERACTION

The app uses a simple, intuitive interface built with Streamlit. The sidebar menu helps users navigate through different sections, including:

1. **Home:** Introduction to the app and its features.
 2. **BMR Calculator:** Calculates BMR and TEE based on user data.
 3. **Calorie Prediction:** Predicts calorie content based on nutritional input.
 4. **Traditional and Smart Recipe Recommendations:** Provides recipe suggestions tailored to the user's nutritional needs.
 5. **Gamification:** Tracks food intake and rewards users.
 6. **Health Goal Tracking:** Assists in maintaining health goals with meal suggestions.
 7. **About:** Provides a brief description of the project.
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5. INTEGRATION WITH WEARABLES

In future iterations, NutriFLEX plans to integrate with wearable devices such as Fitbits and smartwatches to automatically track user activity levels and adjust TEE calculations in real-time. This will enable even more accurate meal planning, as the app will dynamically adjust based on daily physical activity.

Benefits:

- **Increased Accuracy:** By incorporating real-time data from wearables, NutriFLEX will offer **more accurate** calorie recommendations that reflect the user's actual energy needs.
- **Automatic Updates:** Users won't need to manually input their physical activity, as the app will automatically adjust based on wearable data, streamlining the user experience.
- **Holistic Health Tracking:** Integration with wearables will allow NutriFLEX to offer insights not only into nutrition but also overall health, including **fitness and recovery** metrics.

Example Scenario:

- If a user has a particularly active day, NutriFLEX will increase their recommended caloric intake accordingly, ensuring they meet their nutritional needs without over or under-eating. Conversely, if a user has a sedentary day, the app will reduce caloric suggestions to maintain balance.

This **wearable integration** will make **NutriFLEX** a more adaptive and intelligent nutrition assistant, capable of delivering **dynamic, real-time adjustments** that align with each user's unique lifestyle and physical activity levels.

The screenshot displays a dark-themed web interface for the NutriFLEX application. At the top, there are three input fields: 'Enter your desired calorie count:' with a value of 1000, 'Enter your protein intake:' with a value of 'Chia Seeds', and 'Enter allergy ingredient (if any):' with a value of 'Dairy products'. Below these fields is a red-outlined button labeled 'Generate Recipe'. The main content area displays the title 'Mediterranean Quinoa Bowl with Broccoli, Spinach, and Chia Seeds' in white. Below the title, a short description reads: 'This colorful and flavorful bowl is packed with nutrients and provides a healthy dose of protein, fiber, and antioxidants.' Further down, it specifies 'Yields: 1 serving Prep time: 10 minutes Cook time: 20 minutes'. The 'Ingredients:' section lists the following items in a bulleted format: 1 cup quinoa, rinsed; 2 cups vegetable broth; 1 cup broccoli florets; 1 cup spinach, chopped; 1/2 avocado, diced; and 1 tablespoon olive oil.

Figure 6: - ADDED FEATURES

6. ADDED FEATURES

NutriFLEX continues to evolve with additional capabilities for improved nutrition management. Recent updates include:

1. **User-Defined Protein Intake:** The system now allows users to input their desired daily protein intake, which is then factored into the meal and recipe recommendations to meet their nutritional goals.
2. **Allergy Awareness:** A new feature for allergy management lets users specify any ingredients they are allergic to. This input is considered when generating meal plans or recipe suggestions, ensuring the exclusion of those allergens.

Generate AI-based recipe suggestions for specific health conditions

Enter the Health Condition of patient:
diabetes

Current Mood
Stressed

Stress Level (1-10)
1 5 10

Dietary Preferences (e.g., vegan, vegetarian, gluten-free)
gluten-free

Get Meal Suggestions and Recipe

Stress-Busting Salmon with Roasted Asparagus and Quinoa

This recipe is packed with nutrients to help combat stress and is gluten-free, making it a great choice for individuals with diabetes. The salmon provides omega-3 fatty acids, which can help improve mood and reduce inflammation. The asparagus is rich in fiber and antioxidants, while the quinoa is a complete protein and a great source of energy.

Ingredients:

- 1 pound skin-on salmon fillet
- 1 tablespoon olive oil
- 1/2 teaspoon salt
- 1/4 teaspoon black pepper

Figure 7: - SMART HEALTH FEATURE

7. SMART HEALTH FEATURE

The newly introduced *Smart Health* section caters to users with specific health conditions and mental states:

1. **Health Condition Input:** Users can input their past or present health conditions. This data is used by the AI to generate dietary recommendations that align with their medical needs.
2. **Mood-Based Meal Suggestions:** The system factors in the user's mood—whether they are happy, sad, stressed, anxious, or relaxed—when making meal suggestions. The app further refines this with a stress level input (1–10), providing customized recipes that aim to boost mental well-being and physical health.
3. **Dietary Preferences:** Users can now enter their dietary preferences, such as vegan, vegetarian, or gluten-free, which are factored into meal recommendations to better suit individual tastes and health requirements.

Generate a personalized meal plan based on gut microbiome data.

The AI recommends foods to balance the gut flora, support digestion, and improve overall gut health.

Lactobacillus Level
normal

Bifidobacterium Level
high

Firmicutes Level
normal

Bacteroidetes Level
normal

Akkermansia Level
normal

Clostridium difficile Presence
present

Personalized Meal Plan for Improved Gut Health

This meal plan is designed to address your specific microbiome profile, focusing on increasing beneficial bacteria (Lactobacillus and Bifidobacterium), reducing harmful bacteria (Clostridium difficile), and enhancing microbial diversity.

Important Note: This meal plan is for informational purposes only and should not be considered medical advice. Consult with a healthcare professional for personalized guidance and to address any underlying health conditions.

Breakfast:

- **Option 1:** Oatmeal with berries and nuts:
 - 1/2 cup rolled oats cooked in water or unsweetened almond milk
 - 1/4 cup mixed berries (blueberries, raspberries, strawberries)
 - 1 tbsp chopped walnuts or almonds
 - Optional: sprinkle with cinnamon or chia seeds
- **Option 2:** Yogurt parfait with granola and fruit:

Figure 8: - GUT BACTERIA ANALYSIS

8. GUT BACTERIA ANALYSIS

NutriFLEX has also introduced features for generating meal plans based on gut microbiome data, providing highly personalized dietary recommendations that support digestive health:

1. **Microbiome Data Inputs:** Users provide data on their gut bacteria levels (e.g., Lactobacillus, Bifidobacterium, Firmicutes, Bacteroidetes, Akkermansia) and whether Clostridium difficile is present. This microbiome profile is used by the AI to suggest meals that promote a balanced and healthy gut microbiota.
2. **Meal Plan Generation:** The AI generates a full-day meal plan—breakfast, lunch, dinner, and snacks—based on this microbiome data. The plan focuses on enhancing microbial diversity, reducing harmful bacteria, and increasing beneficial bacteria.

9. VOICE ASSISTANCE FEATURE

In future versions of **NutriFLEX**, integrating **Voice Assistance** will enhance user experience by allowing seamless interaction through spoken commands. This feature aims to provide hands-

free navigation and interaction, making it more convenient for users to access nutritional information, calculate metrics like **BMR**, or request personalized **recipe recommendations**.

Example Use Cases:

- *"Hey NutriFLEX, calculate my BMR."*
- *"Suggest a recipe based on 500 calories."*
- *"Track my daily food intake."*

This feature can be implemented using voice recognition APIs, such as Google Speech-to-Text or Amazon Alexa, enabling the app to understand and process spoken language, adding a more interactive and personalized element to **NutriFLEX**.

10. RECIPE INTEGRATION WITH DELIVERY SERVICES

Future versions of NutriFLEX will aim to merge recipes with online food delivery platforms like **Zomato** and **Swiggy**. Users will be able to order ingredients or ready-made meals directly from the app, offering a seamless nutrition solution from planning to execution.

Benefits:

- **Seamless Execution:** Users can effortlessly move from nutrition planning to implementation, saving time and reducing friction in the meal preparation process.
- **Increased Accessibility:** Users who do not have the time or resources to cook can still follow their personalized nutrition plan by ordering healthy, pre-prepared meals.
- **Smart Suggestions:** The app will be able to suggest restaurant meals that match the user's calorie or nutritional needs, offering a smart alternative to home cooking.

Example Scenario:

- After a user receives a **personalized recipe suggestion**, they will have the option to either **order ingredients** for delivery from local grocery stores or order the **ready-made meal** directly from a restaurant on platforms like Zomato or Swiggy.

This integration of recipes with food delivery services will significantly enhance the convenience and functionality of **NutriFLEX**, making it a more comprehensive and user-friendly solution for maintaining a healthy lifestyle.

11. DISCUSSION

The NutriFLEX app successfully integrates machine learning with user interaction to provide personalized nutrition guidance. By allowing users to calculate their BMR, predict calories, and receive tailored meal plans and recipe suggestions, the app empowers individuals to take control of their health. The incorporation of gamification enhances user engagement, encouraging healthier eating habits.

11.1 MODEL PERFORMANCE

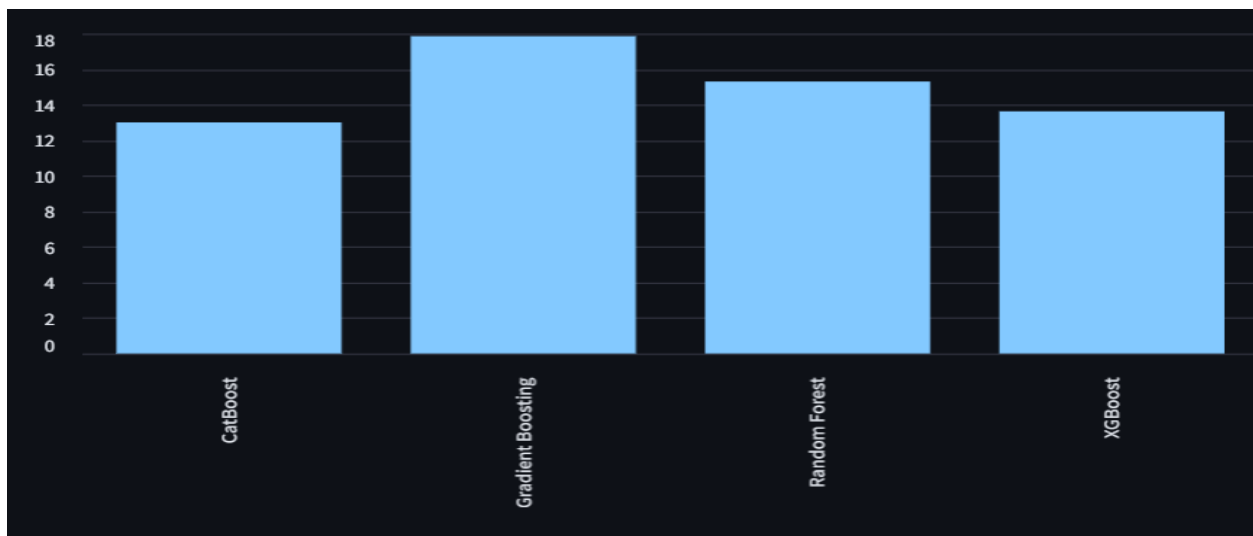


Figure 11.1.A: - Model Performance

R-squared measures the goodness of fit of a regression model. Hence, a higher R-squared indicates the model is a good fit, while a lower R-squared indicates the model is not a good fit.

Model	Mean Square Error (MSE)	R-Square
Random Forest	234.2504	15.30524
XG Boost	185.7045858	13.62735
Gradient Boost	318.8417019	17.85614
Cat Boost	169.1212939	13.00466

Table 11.1.A MODEL PERFORMANCE

Example:

Protein (g)	Fat (g)	Sugar(g)	Copper(mcg)	Iron(mcg)	Predicted Calorie	Model
12.20	3.00	0.40	0.30	0.11	615.54 kcal	Random Forest
					729.87 kcal	Gradient Boosting
					574.11 kcal	Cat Boosting
					621.05 kcal	XGBoost

Table - 11.1.A: - Model Evaluation

Initial performance evaluations of the calorie prediction models showed that **Gradient Boosting** performed slightly better in terms of accuracy compared to the other models. However, user testing with real-world data will be essential to fine-tune the models and ensure reliable predictions.

11.2 USER FEEDBACK

The inclusion of a reward system for food intake, while useful, can be expanded to include more complex challenges, such as weekly health targets or community-based competitions. Feedback from early users suggests the importance of integrating the app with wearables and allowing for more customizable meal plans.

12. FUTURE WORK

The NutriFLEX project is still evolving, and there are several avenues for future development:

1. **Enhanced Gamification:** Expanding the reward system to include achievements, badges, and social sharing to motivate users further.
2. **Wearable Integration:** Incorporating real-time data from fitness trackers and smartwatches for automatic TEE adjustments.
3. **Voice Interface:** Full integration of voice commands for a hands-free experience.
4. **Expanded Recipe Database:** Including more local and international recipes, while also integrating with food delivery services.
5. **Dietary Preferences and Allergies:** Tailoring meal plans and recipes based on dietary restrictions such as gluten-free, vegan, or keto.

CONCLUSION

NutriFLEX stands at the forefront of personalized nutrition by leveraging AI to provide an all-encompassing approach to healthy living. It integrates calorie prediction, tailored meal planning, and smart recipe recommendations while factoring in individual preferences, health conditions, and even mood. The incorporation of gamification not only motivates users to stick to their health goals but also enhances user engagement.

Looking ahead, NutriFLEX will further evolve by offering more specialized features, such as mood-based meal suggestions, microbiome-driven diet plans, and enhanced integration with wearable devices. These future enhancements aim to create an even more dynamic, interactive, and personalized experience, cementing NutriFLEX as a leader in the field of AI-powered nutrition.